The free energy change at the standard state of pH 7 is represented as $\Delta G^{\circ\prime}$. The standard free energy change can be calculated from the change in redox potential $\Delta E^{o\prime}$ of the substrate and products as represented below, where n denotes the number of electrons transferred per mole of reactants, $\Delta E^{o\prime}$ in Volts (V) and $\Delta G^{\circ\prime}$ in kilocalories per mole (kcal mol^{-1}), F is Faraday's constant (96,494°C mol^{-1}). The standard free energy for the reaction is: $\Delta G^{\circ\prime} = -nF\Delta E^{o\prime}$

8.2 ELECTRON TRANSPORT CHAIN

To understand biological oxidation and the electron transport chain, an understanding of oxidation - reduction reactions and the biology of mitochondria are essential. In eukaryotic cells, cellular metabolism takes place in major compartments of the cell. For instance, glycolysis takes place in the cytosol, whereas oxidation of pyruvate, β oxidation of fatty acids and the citric acid cycle take place in the mitochondrial matrix.

The mitochondrial electron transport chain (ETC) is a complicated system, where a series of electron carriers are arranged in the inner membrane of the mitochondria in the order of increasing electron affinity. They transfer the electrons derived from reduced coenzymes to oxygen.

8.2.1 Structure of mitochondria

The size of a mitochondrion is close to the size of a bacterium (ranging between 1 to 2μm). Mitochondria are believed to be developed during an early phase of evolution from aerobic bacteria that entered into symbiosis with primeval anaerobic eukaryotes. This is known as endosymbiosis.

The mitochondria has two membranes, a smooth outer membrane and an inner membrane, which forms numerous folds called as cristae that project into the interior of the mitochondria and enclose the matrix (Fig.8.1). Since the respiratory proteins are embedded in the inner membrane, the density of the cristae is related to the respiratory activity of the cell. For example, the heart muscle with rigorous oxidative metabolism has greater numbers of mitochondria as compared to the liver. Even within one type of tissue, the shape of the mitochondrion varies depending on the functional status of the cell.

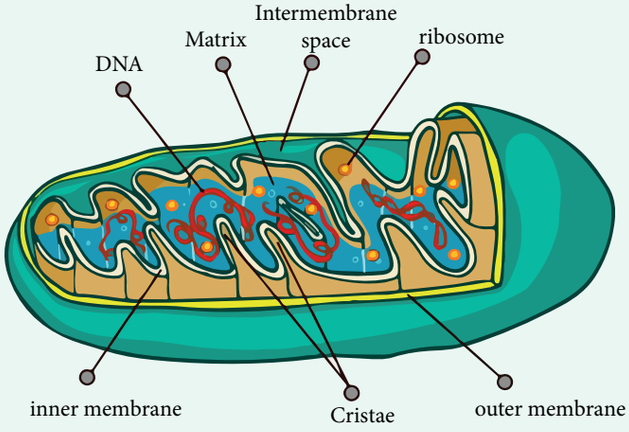


Figure 8.1. Structure of Mitochondria

The mitochondrial membrane is rich in proteins. Porins present in the outermembrane of the mitochondria allow small molecules to be exchanged between the cytoplasm and the inter membrane space. However the inner

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